

EXPERIMENT - 19

1. Problem Statement

The algorithm for calculating the Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA) for 'm' semesters is given below. Each semester has 'n' courses, where 'n' may vary between semesters. Assume that the student has passed all courses. Prepare a test-case table covering valid, invalid, and boundary inputs; write the C program based on the given algorithm; compile and execute it; and test it against the prepared test cases.

Given Algorithm

1. Start.
2. Define the variables and their types: global_id, m, n, semester, course, code, credit, GP, semester_TC, semester_TGC, cumulative_TC, and cumulative_TGC as integers; grade as a character; and SGPA and CGPA as floating-point variables.
3. Get global_id.
4. Validate global_id: If global_id is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
5. Get m.
6. Validate m: If m is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
7. Set the cumulative values: cumulative_TC = 0, cumulative_TGC = 0
8. Repeat Steps 8.1 to 8.8 for m semesters:
 - 8.1. Set the semester totals: semester_TC = 0, semester_TGC = 0
 - 8.2. Get n.
 - 8.3. Validate n: If n is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
 - 8.4. Repeat Steps 8.4.1 to 8.4.9 for n courses:
 - 8.4.1. Get code.
 - 8.4.2. Validate code: If it is not positive, display an error message and request it again. Repeat until a positive value is entered.
 - 8.4.3. Get credit.
 - 8.4.4. Validate credit: If it is not positive, display an error message and request it again. Repeat until a positive value is entered.
 - 8.4.5. Get grade.
 - 8.4.6. Validate grade: Accept only O, A, B, C, D, E, or P. If the grade is invalid, display an error message and request it again. Repeat until a valid grade is entered.
 - 8.4.7. Find GP corresponding to grade using the university's conversion scale:

O = 10, A = 9, B = 8, C = 7, D = 6, E = 5, P = 4

8.4.8. Calculate the semester total grade credits:

$\text{semester_TGC} = \text{semester_TGC} + (\text{GP} * \text{credit})$

8.4.9. Calculate the semester total credits:

$\text{semester_TC} = \text{semester_TC} + \text{credit}$

8.5. Calculate the SGPA of the current semester:

$\text{SGPA} = \text{semester_TGC} / \text{semester_TC}$

8.6. Display semester and SGPA.

8.7. Add the semester grade credits to the cumulative grade credits:

$\text{cumulative_TGC} = \text{cumulative_TGC} + \text{semester_TGC}$

8.8. Add the semester credits to the cumulative credits:

$\text{cumulative_TC} = \text{cumulative_TC} + \text{semester_TC}$

9. Calculate the CGPA:

$\text{CGPA} = \text{cumulative_TGC} / \text{cumulative_TC}$

10. Display global_id, m, cumulative_TC, and CGPA.

11. Stop.

2. Test Case Table

Valid inputs

TC #	Input	Expected Output	Actual Output	Result
TC1	global_id 220101, m = 2 Sem 1: n = 2 → (101, 4, O), (102, 3, A) Sem 2: n = 2 → (201, 3, B), (202, 4, C)	SGPA 1 = 9.57 SGPA 2 = 7.43 Total credits = 14 CGPA = 8.50	SGPA 1 = 9.57 SGPA 2 = 7.43 Total credits = 14 CGPA = 8.50	PASS
TC2	global_id 220102, m = 2, n different Sem 1: n = 3 → (101, 4, O), (102, 3, B), (103, 2, E) Sem 2: n = 1 → (201, 4, A)	SGPA 1 = 8.22 SGPA 2 = 9.00 Total credits = 13 CGPA = 8.46	SGPA 1 = 8.22 SGPA 2 = 9.00 Total credits = 13 CGPA = 8.46	PASS

Boundary inputs

TC #	Input	Expected Output	Actual Output	Result
TC3	Smallest possible values global_id 1, m = 1 Sem 1: n = 1 → (501, 1, O)	SGPA 1 = 10.00 Total credits = 1 CGPA = 10.00	SGPA 1 = 10.00 Total credits = 1 CGPA = 10.00	PASS
TC4	Lowest grade allowed global_id 220104, m = 1 Sem 1: n = 2 → (301, 4, P), (302, 4, P)	SGPA 1 = 4.00 Total credits = 8 CGPA = 4.00	SGPA 1 = 4.00 Total credits = 8 CGPA = 4.00	PASS

Invalid inputs

In every case below the program shows the error message and asks for the same value again. It does not stop and it does not take the wrong value.

TC #	Invalid input given	Expected Output	Actual Output	Result
TC5	global_id = -5, then 220105	Error message, then ask again	Error: Global ID must be positive. Enter again:	PASS
TC6	global_id = -5, then 0, then 220112 (invalid two times)	Error message both the times, then ask again	The error message was shown 2 times, then it accepted 220112	PASS
TC7	m = 0, then 1	Error message, then ask again	Error: number of semesters must be positive. Enter again:	PASS

TC #	Invalid input given	Expected Output	Actual Output	Result
TC8	n = -2, then 1	Error message, then ask again	Error: number of courses must be positive. Enter again:	PASS
TC9	code = 0, then 101	Error message, then ask again	Error: course code must be positive. Enter again:	PASS
TC10	credit = -3, then 3	Error message, then ask again	Error: credits must be positive. Enter again:	PASS
TC11	grade = X, then A	Error message, then ask again	Error: invalid grade. Enter again (O, A, B, C, D, E, P):	PASS
TC12	grade = F, then B	Error message, because F is not in the accepted list	Error: invalid grade. Enter again (O, A, B, C, D, E, P):	PASS

3. Source Code

File name: `sgpa_cgpa.c` (124 lines)

```
/* Program to calculate the Semester Grade Point Average (SGPA) of
'm' semesters and the Cumulative Grade Point Average (CGPA).
Each semester has 'n' courses and 'n' may be different for
every semester. The student has passed all the courses.
The program is written according to the given algorithm. */

#include <stdio.h>

int main(void)
{
    int global_id, m, n, semester, course, code, credit, GP;
    int semester_TC, semester_TGC, cumulative_TC, cumulative_TGC;
    char grade;
    float SGPA, CGPA;

    GP = 0;

    /* Step 3 and Step 4 : get and validate global_id */
    printf("Enter Global ID: ");
    scanf("%d", &global_id);
    while (global_id <= 0) {
        printf("Error: Global ID must be positive. Enter again: ");
        scanf("%d", &global_id);
    }

    /* Step 5 and Step 6 : get and validate m */
    printf("Enter number of semesters: ");
    scanf("%d", &m);
    while (m <= 0) {
        printf("Error: number of semesters must be positive. Enter again: ");
        scanf("%d", &m);
    }

    /* Step 7 : set the cumulative values */
    cumulative_TC = 0;
    cumulative_TGC = 0;

    /* Step 8 : repeat for m semesters */
    for (semester = 1; semester <= m; semester++) {

        /* Step 8.1 : set the semester totals */
        semester_TC = 0;
        semester_TGC = 0;

        /* Step 8.2 and Step 8.3 : get and validate n */
        printf("\nSemester %d\n", semester);
        printf("Enter number of courses: ");
        scanf("%d", &n);
        while (n <= 0) {
            printf("Error: number of courses must be positive. Enter again: ");
            scanf("%d", &n);
        }
    }
}
```

```

/* Step 8.4 : repeat for n courses */
for (course = 1; course <= n; course++) {

    printf(" Course %d\n", course);

    /* Step 8.4.1 and 8.4.2 : get and validate code */
    printf(" Enter course code: ");
    scanf("%d", &code);
    while (code <= 0) {
        printf(" Error: course code must be positive. Enter again: ");
        scanf("%d", &code);
    }

    /* Step 8.4.3 and 8.4.4 : get and validate credit */
    printf(" Enter credits: ");
    scanf("%d", &credit);
    while (credit <= 0) {
        printf(" Error: credits must be positive. Enter again: ");
        scanf("%d", &credit);
    }

    /* Step 8.4.5 and 8.4.6 : get and validate grade */
    printf(" Enter grade (0, A, B, C, D, E, P): ");
    scanf(" %c", &grade);
    while (grade != '0' && grade != 'A' && grade != 'B' &&
        grade != 'C' && grade != 'D' && grade != 'E' &&
        grade != 'P') {
        printf(" Error: invalid grade. Enter again (0, A, B, C, D, E, P): ");
        scanf(" %c", &grade);
    }

    /* Step 8.4.7 : find GP from the grade */
    switch (grade) {
        case '0': GP = 10; break;
        case 'A': GP = 9; break;
        case 'B': GP = 8; break;
        case 'C': GP = 7; break;
        case 'D': GP = 6; break;
        case 'E': GP = 5; break;
        case 'P': GP = 4; break;
    }

    /* Step 8.4.8 and 8.4.9 : add to the semester totals */
    semester_TGC = semester_TGC + (GP * credit);
    semester_TC = semester_TC + credit;
}

/* Step 8.5 : calculate the SGPA of this semester */
SGPA = (float) semester_TGC / semester_TC;

/* Step 8.6 : display semester and SGPA */
printf("\nSemester %d : SGPA = %.2f\n", semester, SGPA);

/* Step 8.7 and 8.8 : add to the cumulative values */
cumulative_TGC = cumulative_TGC + semester_TGC;
cumulative_TC = cumulative_TC + semester_TC;
}

/* Step 9 : calculate the CGPA */
CGPA = (float) cumulative_TGC / cumulative_TC;

```

```
/* Step 10 : display the final result */
printf("\n----- RESULT ----- \n");
printf("Global ID          : %d\n", global_id);
printf("Number of semesters : %d\n", m);
printf("Total credits       : %d\n", cumulative_TC);
printf("CGPA                : %.2f\n", CGPA);

/* Step 11 : stop */
return 0;
}
```

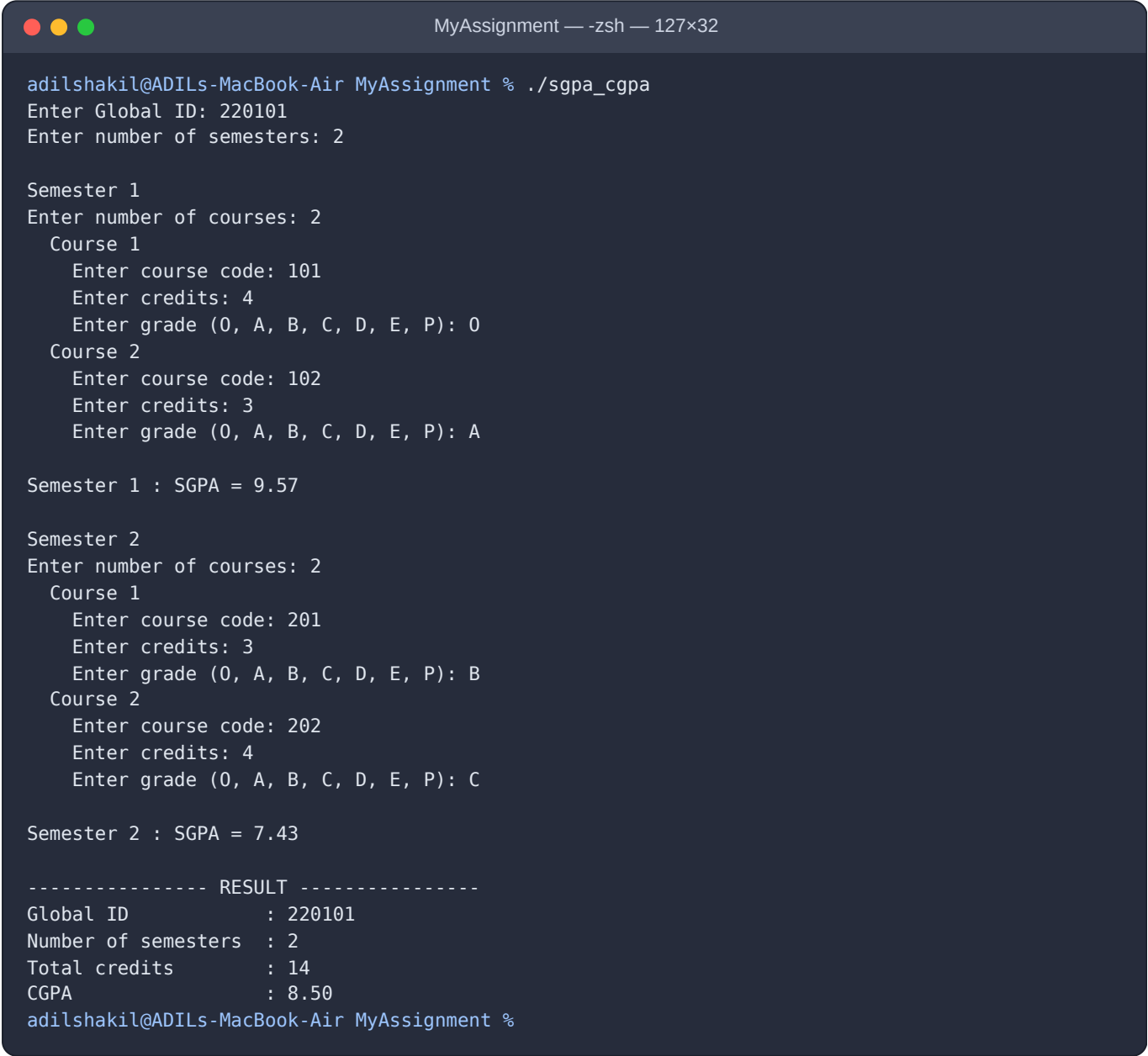
4. Compilation and Output

I wrote the program in the `MyAssignment` folder using the `nano` editor, then compiled and ran it:

```
$ cd ~/Desktop/MyAssignment
$ nano sgpa_cgpa.c
$ gcc sgpa_cgpa.c -o sgpa_cgpa
$ ./sgpa_cgpa
```

The program compiled in the first attempt itself, without any error or warning.

TC1 : two semesters with two courses each



```
MyAssignment — zsh — 127x32

adilshakil@ADILs-MacBook-Air MyAssignment % ./sgpa_cgpa
Enter Global ID: 220101
Enter number of semesters: 2

Semester 1
Enter number of courses: 2
  Course 1
    Enter course code: 101
    Enter credits: 4
    Enter grade (0, A, B, C, D, E, P): 0
  Course 2
    Enter course code: 102
    Enter credits: 3
    Enter grade (0, A, B, C, D, E, P): A

Semester 1 : SGPA = 9.57

Semester 2
Enter number of courses: 2
  Course 1
    Enter course code: 201
    Enter credits: 3
    Enter grade (0, A, B, C, D, E, P): B
  Course 2
    Enter course code: 202
    Enter credits: 4
    Enter grade (0, A, B, C, D, E, P): C

Semester 2 : SGPA = 7.43

----- RESULT -----
Global ID      : 220101
Number of semesters : 2
Total credits  : 14
CGPA           : 8.50
adilshakil@ADILs-MacBook-Air MyAssignment %
```

TC2 : number of courses is different in the two semesters

```
MyAssignment — -zsh — 127×32

adilshakil@ADILs-MacBook-Air MyAssignment % ./sgpa_cgpa
Enter Global ID: 220102
Enter number of semesters: 2

Semester 1
Enter number of courses: 3
  Course 1
    Enter course code: 101
    Enter credits: 4
    Enter grade (0, A, B, C, D, E, P): 0
  Course 2
    Enter course code: 102
    Enter credits: 3
    Enter grade (0, A, B, C, D, E, P): B
  Course 3
    Enter course code: 103
    Enter credits: 2
    Enter grade (0, A, B, C, D, E, P): E

Semester 1 : SGPA = 8.22

Semester 2
Enter number of courses: 1
  Course 1
    Enter course code: 201
    Enter credits: 4
    Enter grade (0, A, B, C, D, E, P): A

Semester 2 : SGPA = 9.00

----- RESULT -----
Global ID      : 220102
Number of semesters : 2
Total credits   : 13
CGPA           : 8.46
adilshakil@ADILs-MacBook-Air MyAssignment %
```

TC3 and TC4 : boundary cases

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./sgpa_cgpa
```

```
Enter Global ID: 1
```

```
Enter number of semesters: 1
```

```
Semester 1
```

```
Enter number of courses: 1
```

```
Course 1
```

```
Enter course code: 501
```

```
Enter credits: 1
```

```
Enter grade (0, A, B, C, D, E, P): 0
```

```
Semester 1 : SGPA = 10.00
```

```
----- RESULT -----
```

```
Global ID      : 1
```

```
Number of semesters : 1
```

```
Total credits   : 1
```

```
CGPA            : 10.00
```

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./sgpa_cgpa
```

```
Enter Global ID: 220104
```

```
Enter number of semesters: 1
```

```
Semester 1
```

```
Enter number of courses: 2
```

```
Course 1
```

```
Enter course code: 301
```

```
Enter credits: 4
```

```
Enter grade (0, A, B, C, D, E, P): P
```

```
Course 2
```

```
Enter course code: 302
```

```
Enter credits: 4
```

```
Enter grade (0, A, B, C, D, E, P): P
```

```
Semester 1 : SGPA = 4.00
```

```
----- RESULT -----
```

```
Global ID      : 220104
```

```
Number of semesters : 1
```

```
Total credits   : 8
```

```
CGPA            : 4.00
```

```
adilshakil@ADILs-MacBook-Air MyAssignment %
```

TC5 and TC6 : invalid global_id, asked again until it is positive

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./sgpa_cgpa
Enter Global ID: -5
Error: Global ID must be positive. Enter again: 220105
Enter number of semesters: 1
```

```
Semester 1
Enter number of courses: 1
  Course 1
    Enter course code: 101
    Enter credits: 3
    Enter grade (0, A, B, C, D, E, P): A
```

```
Semester 1 : SGPA = 9.00
```

```
----- RESULT -----
Global ID      : 220105
Number of semesters : 1
Total credits   : 3
CGPA           : 9.00
```

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./sgpa_cgpa
Enter Global ID: -5
Error: Global ID must be positive. Enter again: 0
Error: Global ID must be positive. Enter again: 220112
Enter number of semesters: 1
```

```
Semester 1
Enter number of courses: 1
  Course 1
    Enter course code: 101
    Enter credits: 3
    Enter grade (0, A, B, C, D, E, P): A
```

```
Semester 1 : SGPA = 9.00
```

```
----- RESULT -----
Global ID      : 220112
Number of semesters : 1
Total credits   : 3
CGPA           : 9.00
```

```
adilshakil@ADILs-MacBook-Air MyAssignment %
```

TC7 to TC10 : invalid m, n, code and credit

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./sgpa_cgpa
Enter Global ID: 220106
Enter number of semesters: 0
Error: number of semesters must be positive. Enter again: 1

Semester 1
Enter number of courses: -2
Error: number of courses must be positive. Enter again: 1
  Course 1
    Enter course code: 0
    Error: course code must be positive. Enter again: 101
    Enter credits: -3
    Error: credits must be positive. Enter again: 3
    Enter grade (0, A, B, C, D, E, P): A

Semester 1 : SGPA = 9.00

----- RESULT -----
Global ID       : 220106
Number of semesters : 1
Total credits    : 3
CGPA            : 9.00
adilshakil@ADILs-MacBook-Air MyAssignment %
```

TC11 and TC12 : invalid grade

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./sgpa_cgpa
```

```
Enter Global ID: 220110
```

```
Enter number of semesters: 1
```

```
Semester 1
```

```
Enter number of courses: 1
```

```
Course 1
```

```
Enter course code: 101
```

```
Enter credits: 3
```

```
Enter grade (0, A, B, C, D, E, P): X
```

```
Error: invalid grade. Enter again (0, A, B, C, D, E, P): A
```

```
Semester 1 : SGPA = 9.00
```

```
----- RESULT -----
```

```
Global ID      : 220110
```

```
Number of semesters : 1
```

```
Total credits   : 3
```

```
CGPA            : 9.00
```

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./sgpa_cgpa
```

```
Enter Global ID: 220111
```

```
Enter number of semesters: 1
```

```
Semester 1
```

```
Enter number of courses: 1
```

```
Course 1
```

```
Enter course code: 101
```

```
Enter credits: 3
```

```
Enter grade (0, A, B, C, D, E, P): F
```

```
Error: invalid grade. Enter again (0, A, B, C, D, E, P): B
```

```
Semester 1 : SGPA = 8.00
```

```
----- RESULT -----
```

```
Global ID      : 220111
```

```
Number of semesters : 1
```

```
Total credits   : 3
```

```
CGPA            : 8.00
```

```
adilshakil@ADILs-MacBook-Air MyAssignment %
```